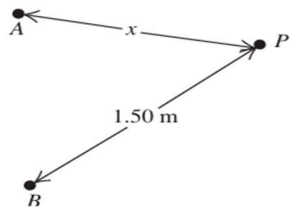
Homework 08 with solutions

1. Two small stereo speakers A and B that are 1.40 m apart are sending out sound of wavelength 34 cm in all directions and all in phase. A person at point P starts out equidistant from both speakers and walks so that he is always 1.50 m from speaker B (shown in fig). For what values of x will the sound this person hears be (a) maximally reinforced, (b) cancelled? Limit your solution to the cases where $x \leq 1.50m$.



Solution

IDENTIFY: The sound will be maximally reinforced when the path difference is an integral multiple of wavelengths and cancelled when it is an odd number of half wavelengths.
SET UP: Constructive interference occurs for $r_2 - r_1 = m\lambda,\ m = 0, \pm 1, \pm 2, \ldots$. Destructive interference occurs for $r_2 - r_1 = (m + \frac{1}{2})\lambda, m = 0, \pm 1, \pm 2 \ldots$. For this problem, $r_2 = 150$ cm and $r_1 = x$. The path taken by the person ensures that x is in the range $0 \leq x \leq 150$ cm.
EXECUTE: **(a)** $150\text{ cm} - x = m(34\text{ cm})$. $x = 150\text{ cm} - m(34\text{ cm})$. For $m = 0, 1, 2, 3, 4$ the values of x are 150 cm, 116 cm, 82 cm, 48 cm, 14 cm.
(b) $150\text{ cm} - x = (m + \frac{1}{2})(34\text{ cm})$. $x = 150\text{ cm} - (m + \frac{1}{2})(34\text{ cm})$. For $m = 0, 1, 2, 3$ the values of x are 133 cm, 99 cm, 65 cm, 31 cm.
EVALUATE: When $x = 116$ cm the path difference is $150\text{ cm} - 116\text{ cm} = 34$ cm, which is one wavelength. When $x = 133$ cm the path difference is 17 cm, which is one-half wavelength.

2. Two speakers that are 15.0 m apart produce in-phase sound waves of frequency 250.0 Hz in a room where the speed of sound is 340.0 m/s. A woman starts out at the midpoint between the two speakers. The room's walls and ceiling are covered with absorbers to eliminate reflections, and she listens with only one ear for best precision. (a) What does she hear: constructive or destructive interference? Why? (b) She now walks slowly toward one of the speakers. How far from the center must she walk before she first hears the sound reach a minimum intensity? (c) How far from the center must she walk before she first hears the sound maximally enhanced?

Solution

IDENTIFY: The sound will be maximally reinforced when the path difference is an integral multiple of wavelengths and cancelled when it is an odd number of half wavelengths.

SET UP: When she is at the midpoint between the two speakers the path difference $r_2 - r_1$ is zero. When she walks a distance d toward one speaker, r_2 increases by d and r_1 decreases by d, so the path difference changes by $2d$. Path difference $= m\lambda$ $(m = 0, \pm 1, \pm 2, \ldots)$ gives constructive interference and path difference $= (m + \frac{1}{2})\lambda$ $(m = 0, \pm 1, \pm 2, \ldots)$ gives destructive interference.

EXECUTE: $\lambda = \dfrac{v}{f} = \dfrac{340.0 \text{ m/s}}{250.0 \text{ Hz}} = 1.36 \text{ m}.$

(a) The path difference is zero, so the interference is constructive.

(b) Destructive interference occurs, so the path difference equals $\lambda/2$. $2d = \dfrac{\lambda}{2}$ which gives

$$d = \frac{\lambda}{4} = \frac{1.36 \text{ m}}{4} = 34.0 \text{ cm}.$$

(c) Constructive interference occurs, so the path difference equals λ. $2d = \lambda$ which gives

$$d = \frac{\lambda}{2} = \frac{1.36 \text{ m}}{2} = 68.0 \text{ cm}.$$

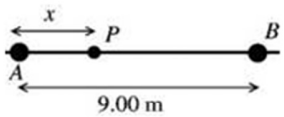
EVALUATE: If she keeps walking, she will possibly find additional places where constructive and destructive interference occur.

3. A radio transmitting station operating at a frequency of 120 MHz has two identical antennas that radiate in phase. Antenna B is 9.00 m to the right of antenna A. Consider point P between the antennas and along the line connecting them, a horizontal distance x to the right of antenna A. For what values of x will constructive interference occur at point P?

Solution

IDENTIFY: Use $c = f\lambda$ to calculate the wavelength of the transmitted waves. Compare the difference in the distance from A to P and from B to P. For constructive interference this path difference is an integer multiple of the wavelength.

SET UP: Consider Figure 35.5.



The distance of point P from each coherent source is $r_A = x$ and $r_B = 9.00\text{ m} - x$.

Figure 35.5

EXECUTE: The path difference is $r_B - r_A = 9.00\text{ m} - 2x$.

$r_B - r_A = m\lambda,\ m = 0,\ \pm 1, \pm 2,\ \ldots$

$$\lambda = \frac{c}{f} = \frac{2.998\times 10^8\text{ m/s}}{120\times 10^6\text{ Hz}} = 2.50\text{ m}$$

Thus $9.00\text{ m} - 2x = m(2.50\text{ m})$ and $x = \dfrac{9.00\text{ m} - m(2.50\text{ m})}{2} = 4.50\text{ m} - (1.25\text{ m})m$. x must lie in the range 0 to 9.00 m since P is said to be between the two antennas.

$m = 0$ gives $x = 4.50$ m
$m = +1$ gives $x = 4.50\text{ m} - 1.25\text{ m} = 3.25$ m
$m = +2$ gives $x = 4.50\text{ m} - 2.50\text{ m} = 2.00$ m
$m = +3$ gives $x = 4.50\text{ m} - 3.75\text{ m} = 0.75$ m
$m = -1$ gives $x = 4.50\text{ m} + 1.25\text{ m} = 5.75$ m
$m = -2$ gives $x = 4.50\text{ m} + 2.50\text{ m} = 7.00$ m
$m = -3$ gives $x = 4.50\text{ m} + 3.75\text{ m} = 8.25$ m

All other values of m give values of x out of the allowed range. Constructive interference will occur for $x = 0.75$ m, 2.00 m, 3.25 m, 4.50 m, 5.75 m, 7.00 m and 8.25 m.

EVALUATE: Constructive interference occurs at the midpoint between the two sources since that point is the same distance from each source. The other points of constructive interference are symmetrically placed relative to this point.

4. Young's experiment is performed with light from excited helium atoms ($\lambda = 502nm$). Fringes are measured carefully on a screen 1.20 m away from the double slit, and the center of the 20th fringe (not counting the central bright fringe) is found to be 10.6 mm from the center of the central bright fringe. What is the separation of the two slits?

Solution

IDENTIFY: The value of y_{20} is much smaller than R and the approximate expression $y_m = R\dfrac{m\lambda}{d}$ is accurate.

SET UP: $y_{20} = 10.6\times 10^{-3}$ m.

EXECUTE: $d = \dfrac{20R\lambda}{y_{20}} = \dfrac{(20)(1.20\text{ m})(502\times 10^{-9}\text{ m})}{10.6\times 10^{-3}\text{ m}} = 1.14\times 10^{-3}\text{ m} = 1.14\text{ mm}$

EVALUATE: $\tan\theta_{20} = \dfrac{y_{20}}{R}$ so $\theta_{20} = 0.51°$ and the approximation $\sin\theta_{20} \approx \tan\theta_{20}$ is very accurate.

5. Two slits spaced 0.450 mm apart are placed 75.0 cm from a screen. What is the distance between the second and third dark lines of the interference pattern on the screen when the slits are illuminated with coherent light with a wavelength of 500 nm?

Solution

IDENTIFY and **SET UP:** The dark lines correspond to destructive interference and hence are located by Eq. (35.5):

$$d\sin\theta = \left(m+\frac{1}{2}\right)\lambda \text{ so } \sin\theta = \frac{\left(m+\frac{1}{2}\right)\lambda}{d},\ m = 0, \pm 1, \pm 2, \ldots$$

Solve for θ that locates the second and third dark lines. Use $y = R\tan\theta$ to find the distance of each of the dark lines from the center of the screen.

EXECUTE: 1st dark line is for $m = 0$

2nd dark line is for $m = 1$ and $\sin\theta_1 = \frac{3\lambda}{2d} = \frac{3(500\times10^{-9}\text{ m})}{2(0.450\times10^{-3}\text{m})} = 1.667\times10^{-3}$ and $\theta_1 = 1.667\times10^{-3}$ rad

3rd dark line is for $m = 2$ and $\sin\theta_2 = \frac{5\lambda}{2d} = \frac{5(500\times10^{-9}\text{ m})}{2(0.450\times10^{-3}\text{m})} = 2.778\times10^{-3}$ and $\theta_2 = 2.778\times10^{-3}$ rad

(Note that θ_1 and θ_2 are small so that the approximation $\theta \approx \sin\theta \approx \tan\theta$ is valid.) The distance of each dark line from the center of the central bright band is given by $y_m = R\tan\theta$, where $R = 0.850$ m is the distance to the screen.

$\tan\theta \approx \theta$ so $y_m = R\theta_m$

$y_1 = R\theta_1 = (0.750\text{ m})(1.667\times10^{-3}\text{ rad}) = 1.25\times10^{-3}\text{ m}$

$y_2 = R\theta_2 = (0.750\text{ m})(2.778\times10^{-3}\text{ rad}) = 2.08\times10^{-3}\text{ m}$

$\Delta y = y_2 - y_1 = 2.08\times10^{-3}\text{ m} - 1.25\times10^{-3}\text{ m} = 0.83\text{ mm}$

EVALUATE: Since θ_1 and θ_2 are very small we could have used Eq. (35.6), generalized to destructive interference: $y_m = R\left(m+\frac{1}{2}\right)\lambda/d$.

6. Coherent light with wavelength 600 nm passes through two very narrow slits and the interference pattern is observed on a screen 3.00 m from the slits. The first-order bright fringe is at 4.84 mm from the center of the central bright fringe. For what wavelength of light will the first-order dark fringe be observed at this same point on the screen?

Solution

IDENTIFY and **SET UP:** Use the information given about the bright fringe to find the distance d between the two slits. Then use Eq. (35.5) and $y = R\tan\theta$ to calculate λ for which there is a first-order dark fringe at this same place on the screen.

EXECUTE: $y_1 = \dfrac{R\lambda_1}{d}$, so $d = \dfrac{R\lambda_1}{y_1} = \dfrac{(3.00\text{ m})(600\times10^{-9}\text{ m})}{4.84\times10^{-3}\text{ m}} = 3.72\times10^{-4}$ m. (R is much greater than d, so Eq. 35.6 is valid.) The dark fringes are located by $d\sin\theta = \left(m+\dfrac{1}{2}\right)\lambda,\ m = 0, \pm1, \pm2, \ldots$ The first-order dark fringe is located by $\sin\theta = \lambda_2/2d$, where λ_2 is the wavelength we are seeking.

$$y = R\tan\theta \approx R\sin\theta = \frac{\lambda_2 R}{2d}$$

We want λ_2 such that $y = y_1$. This gives $\dfrac{R\lambda_1}{d} = \dfrac{R\lambda_2}{2d}$ and $\lambda_2 = 2\lambda_1 = 1200$ nm.

EVALUATE: For $\lambda = 600$ nm the path difference from the two slits to this point on the screen is 600 nm. For this same path difference (point on the screen) the path difference is $\lambda/2$ when $\lambda = 1200$ nm.

7. In a two-slit interference pattern, the intensity at the peak of the central maximum is I_0 (a) At a point in the pattern where the phase difference between the waves from the two slits is 60.0, what is the intensity? (b) What is the path difference for 480-nm light from the two slits at a point where the phase angle is 60.0°?

Solution

IDENTIFY: Eq. (35.10): $I = I_0\cos^2(\phi/2)$. Eq. (35.11): $\phi = (2\pi/\lambda)(r_2 - r_1)$.

SET UP: ϕ is the phase difference and $(r_2 - r_1)$ is the path difference.

EXECUTE: **(a)** $I = I_0(\cos 30.0°)^2 = 0.750 I_0$

(b) $60.0° = (\pi/3)$ rad. $(r_2 - r_1) = (\phi/2\pi)\lambda = [(\pi/3)/2\pi]\lambda = \lambda/6 = 80$ nm.

EVALUATE: $\phi = 360°/6$ and $(r_2 - r_1) = \lambda/6$.

8. Coherent light with wavelength 500 nm passes through narrow slits separated by 0.340 mm. At a distance from the slits large compared to their separation, what is the phase difference (in radians) in the light from the two slits at an angle of from the centerline?

Solution

IDENTIFY and **SET UP:** The phase difference ϕ is given by $\phi = (2\pi d/\lambda)\sin\theta$ (Eq. 35.13.)

EXECUTE: $\phi = [2\pi(0.340\times10^{-3}\text{ m})/(500\times10^{-9}\text{ m})\sin 23.0° = 1670$ rad

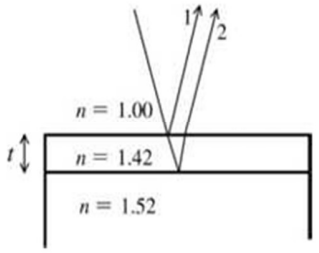
EVALUATE: The mth bright fringe occurs when $\phi = 2\pi m$, so there are a large number of bright fringes within 23.0° from the centerline. Note that Eq. (35.13) gives ϕ in radians.

9. What is the thinnest film of a coating with $n = 1.42$ on glass ($n = 1.52$) for which destructive interference of the red component (650 nm) of an incident white light beam in air can take place by reflection?

Solution

IDENTIFY: Consider interference between rays reflected at the upper and lower surfaces of the film. Consider phase difference due to the path difference of $2t$ and any phase differences due to phase changes upon reflection.

SET UP: Consider Figure 35.25.



Both rays (1) and (2) undergo a $180°$ phase change on reflection, so there is no net phase difference introduced and the condition for destructive interference is $2t = \left(m + \frac{1}{2}\right)\lambda$.

Figure 35.25

EXECUTE: $t = \dfrac{\left(m + \frac{1}{2}\right)\lambda}{2}$; thinnest film says $m = 0$ so $t = \dfrac{\lambda}{4}$.

$$\lambda = \frac{\lambda_0}{1.42} \text{ and } t = \frac{\lambda_0}{4(1.42)} = \frac{650 \times 10^{-9}\ \text{m}}{4(1.42)} = 1.14 \times 10^{-7}\ \text{m} = 114\ \text{nm}$$

EVALUATE: We compared the path difference to the wavelength in the film, since that is where the path difference occurs.

10. A plate of glass 9.00 cm long is placed in contact with a second plate and is held at a small angle with it by a metal strip 0.0800 mm thick placed under one end. The space between the plates is filled with air. The glass is illuminated from above with light having a wavelength in air of 656 nm. How many interference fringes are observed per centimeter in the reflected light?

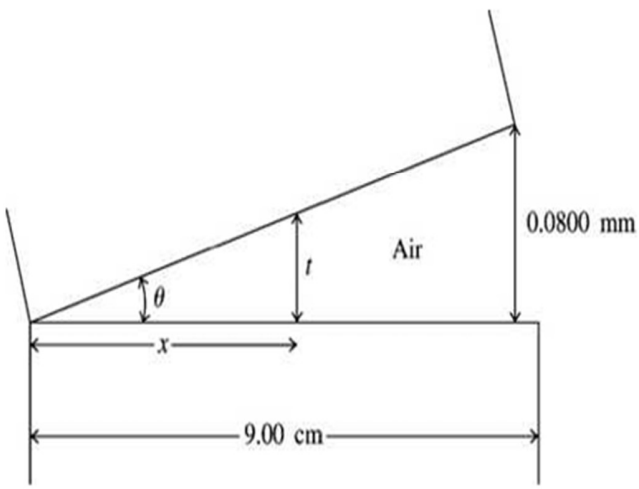
Solution

IDENTIFY: The fringes are produced by interference between light reflected from the top and from the bottom surfaces of the air wedge. The refractive index of glass is greater than that of air, so the waves reflected from the top surface of the air wedge have no reflection phase shift and the waves reflected from the bottom surface of the air wedge do have a half-cycle reflection phase shift. The condition for constructive interference (bright fringes) therefore is $2t = (m+\frac{1}{2})\lambda$.

SET UP: The geometry of the air wedge is sketched in Figure 35.28.

EXECUTE: $\tan\theta = \dfrac{0.0800\text{ mm}}{90.0\text{ mm}} = 8.89\times10^{-4}$. $\tan\theta = \dfrac{t}{x}$ so $t = (8.89\times10^{-4})x$. $t_m = (m+\frac{1}{2})\dfrac{\lambda}{2}$.

$x_m = (m+\frac{1}{2})\dfrac{\lambda}{2(8.89\times10^{-4})}$ and $x_{m+1} = (m+\frac{3}{2})\dfrac{\lambda}{2(8.89\times10^{-4})}$. The distance along the plate between adjacent fringes is $\Delta x = x_{m+1} - x_m = \dfrac{\lambda}{2(8.89\times10^{-4})} = \dfrac{656\times10^{-9}\text{ m}}{2(8.89\times10^{-4})} = 3.69\times10^{-4}\text{ m} = 0.369\text{ mm}$.

The number of fringes per cm is $\dfrac{1.00}{\Delta x} = \dfrac{1.00}{0.0369\text{ cm}} = 27.1$ fringes/cm.

EVALUATE: As $t \to 0$ the interference is destructive and there is a dark fringe at the line of contact between the two plates.